



Thermal transparency of tunable thermal metamaterials composed of composite cylinder or sphere with imperfect interface

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ABSTRACT

We present an exact analysis of tunable thermal metamaterials made of composite cylinder or sphere with imperfect interface effects. The composite cylinder or sphere is composed of an isotropic core coated with a layer of curvilinearly anisotropic material. The interface between the core and the anisotropic layer could be imperfectly bonded, either with low-conductivity or high-conductivity type of interface. Based on an effective medium theory of neutral inclusion, we show that it is possible to balance in different ways among the material and geometric parameters, together with bonding parameters, so that the composite cylinder or sphere is thermally transparent to the background medium. We show how these parameters will affect the thermal intensity within our targeted region, either greatly enhanced or shielded. Remarkably, we show that there exists an interesting link between the core conductivity and the imperfect interface parameter for either type of imperfect interface, highlighting how the effect of core conductivity can be correlated exactly to the effect of bonding imperfectness. We also validate our analytic results with numerical simulations based on finite element calculations. Our findings provide insightful guidance towards the understanding of thermal transport mechanism across imperfect interfaces in thermal metamaterials.

1. Introduction

Heat flow diffuses from a region of high temperature to a region of low temperature. According to the Fourier law, on a macroscopic scale, the temperature is governed by $\nabla \cdot (\mathbf{k} \nabla T) = 0$ under steady-state conditions without heat source, where $\mathbf{k}$ is the thermal conductivity with a dimension of $\text{W} \cdot \text{m}^{-1} \text{K}^{-1}$ and T is the temperature. For a given temperature gradient, the heat flow rate and direction will be controlled by the thermal conductivity tensor $\mathbf{k}$ of the medium. Specifically, the magnitude of conductivity tensor controls the rate of heat flow, while the tensorial form $\mathbf{k}$ will affect the flow direction in the medium.

Significant progress has been made in the design of thermal metamaterials to control the flow of heat. Via a proper design of thermal metamaterials, thermal intensity within a certain area can be greatly manipulated, either enhanced, shielded or distorted [1,2]. Numerous works have examined the subject in different aspects [3–12] broadly. In addition, substantial progress has been made in experimental implementations [13–23]. For a general exposition of the context, one can refer to the review articles [24–31] and the monographs [32,33]. Most of these studies assumed perfect bonding along interfaces. In small-

length scales and some other situations, an ideal interface framework could not be adequate to describe the physical behavior [34,35]. In the literature, researches on interface effects of thermal metamaterials are not many. Among relevant studies, Zheng and Li [36] demonstrated that, when the interface is not perfectly ideal, the temperature field outside an ideal cloak could be greatly distorted. More recently, Chen and Lin [37,38] examined the effects of imperfect interfaces of a circular ring or spherical shell. They demonstrated that the effect of imperfect interface will significantly affect the energy flow across the interface, and thereby greatly influenced the thermal intensity within the targeted area.

In this study, we consider the configurations of composite cylinder and composite sphere. The idea is to explore how the core material will affect the performance of thermal metamaterial. It turns out that the core region will play a crucial role in the design of metamaterial and could have wide implications. Specifically, we consider a circular or spherical core inclusion surrounded by a curvilinearly anisotropic layer. We consider that the interface between the core and the anisotropic layer can be imperfectly bonded. The interface can be either of low-conductivity (LC) or high-conductivity (HC) type, or just perfectly

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bonded. The remaining medium, outside the composite cylinder or sphere, is the background medium with a certain isotropic conductivity. We show that this composite cylinder or sphere can be utilized as a functional device to enhance or to shield thermal intensity within the core region. Our objective is to first explore how the core material, together with the curvilinearly anisotropic material, counting into interface imperfectness, can be suitably balanced so that the thermal intensity field in the background medium will not be distorted, as if the whole medium is homogeneous throughout. This situation is referred to as thermal transparency, or thermal invisibility, as in the context of composite mechanics for neutral inclusions [39,40]. The second objective is to illustrate how these material and geometric parameters, together with bonding parameters, will affect the thermal intensity within our targeted region.

Remarkably, we find that an exact connection exists between the core conductivity and the imperfect interface parameters for both types of imperfect interfaces. This exact link provides a novel physical information, highlighting how the core conductivity will be able to compensate the effect of bonding imperfectness. For thermal transparency conditions, the effect of normalized core conductivity and normalized imperfect interface parameters can be correlated exactly. In addition, we find also that the thermal intensity within our targeted region will be highly related to the core conductivity and to the perfectness of bonding condition. All these findings are derived in simple analytic expressions. In simplified situations, we demonstrate that our analytic expressions can be exactly reduced to previously known results. Numerical illustrations on the thermal transparency conditions and on the thermal intensity within our targeted area will be presented, which will exemplify how the core conductivity will affect the transparency behavior. To be feasible in practice, numerical simulations based on finite element calculations are conducted, in which the imperfect bonding interface is modeled as a thin interphase with either very low or very high conductivity, and the curvilinearly anisotropic layer is simulated by a layered medium with two different isotropic phases. The calculations is verified with our analytical solutions.

For simplicity of exposition, we start from a summary of our main findings. These include analytic expressions of thermal transparency conditions for systems of two- or three-dimensional configurations with either type of imperfect interfaces. Thermal intensity within the targeted area versus a uniform prescribed intensity is also derived in closed forms. A description of physical interpretations and mathematical framework of LC- and HC-type interfaces will be outlined in Section 2. The main theoretical formulations will be presented in Section 3 and an appendix is included for detailed derivations. Numerical illustrations on

the effects of LC-type and HC-type interface will be presented in full spectrum, so that it is easy to perceive the opposite trend of both types of interfaces and that the perfect bonding situation stands just in the middle point between those of LC- and HC- type interfaces. Lastly, a concluding remark will be addressed in Section 5. Our analytic findings together with numerical illustrations will offer a general perception that could add to the understanding of heat transport mechanism across imperfect interfaces in thermal metamaterials and will provide design strategies for the control and management of thermal metamaterials with imperfect bonding interfaces.

2. Main results

We consider a circular or spherical core region (Region I) surrounded by an anisotropic layer (Region II), as depicted in Fig. 1. Region III is the surrounding background material with isotropic conductivity $k = k_0$. We consider that the interface between the inclusion and the anisotropic layer could be imperfectly bonded. Region I is isotropic with conductivity $k = k_1$, which is the targeted region for heat flux control. Region II is a homogeneous anisotropic layer with constant radial and circumferential (or tangential) conductivities,

$$\mathbf{k} = \begin{pmatrix} k_r & 0 \\ 0 & k_\theta \end{pmatrix}, \quad (1)$$

for a two-dimensional configuration, or

$$\mathbf{k} = \begin{pmatrix} k_r & 0 & 0 \\ 0 & k_t & 0 \\ 0 & 0 & k_t \end{pmatrix}, \quad (2)$$

for a three-dimensional configuration. The interface between Region I and Region II can be either of low-conductivity or high-conductivity type, or just perfectly bonded. Along a certain direction, a uniform thermal intensity E is prescribed.

To begin with, let us first recapitulate briefly previous findings for simple expositions. Some years ago, Han et al. [4] and Chen et al. [8] showed that a circular ring or a spherical shell can act as a thermal device of invisibility cloak or concentrator. They demonstrated an anisotropic circular ring or shell can be placed inside an isotropic background medium without distorting the uniform intensity field. Apparently, if the circular ring or shell is isotropic, it must be the same with the background material. However, if the circular ring or spherical shell is curvilinearly anisotropic, as in (1) or (2) with different conductivities in the radial and circumferential (or tangential) directions, it appears that it is indeed possible to balance the difference between the

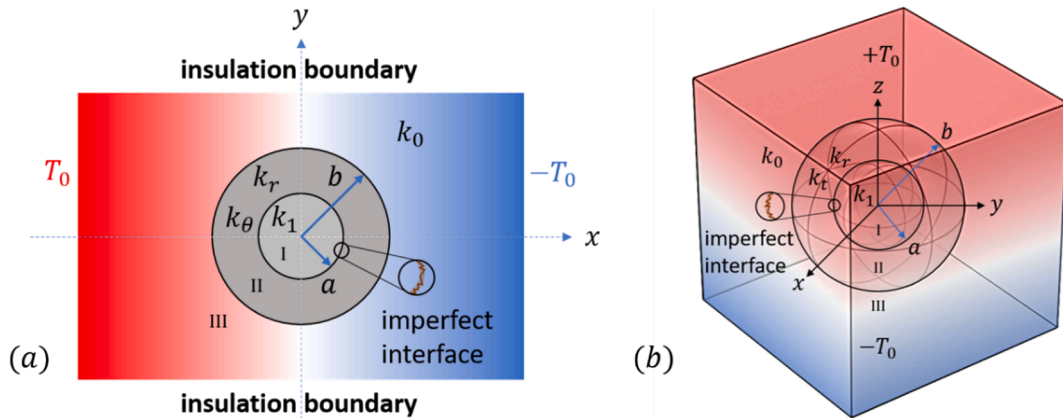


Fig. 1. Schematic diagrams of the configurations of composite cylinder (Fig. 1a) and composite sphere (Fig. 1b). The core region $r \leq a$ is Region I with thermal conductivity $k = k_1$, serving as the targeted area for controlling heat flux. The region $a \leq r \leq b$ is Region II, with thermal conductivities $\mathbf{k}$, given in Eq.(1) or Eq.(2), as the anisotropic cloak. The region $r \geq b$ is the background material, labelled as Region III, with thermal conductivity $k = k_0$, influenced by Regions I and II. The interface at $r = a$ could have imperfect contact, with either LC- or HC-type imperfect interface.

two conductivities with that of the background material. The exact results are simple with the relation

$$k_r k_\theta = k_0^2, \quad (3)$$

for a circular ring [4], and

$$k_r(2k_t - k_0) = k_0^2, \quad (4)$$

for a spherical shell [8]. Equations (3) and (4) can be referred to as the thermal transparency or invisibility condition. In particular, the thermal intensity inside the ring or shell can be greatly enhanced or shielded by varying the anisotropic material parameters in (1) or (2). Chen and Lin [37,38] later showed that, with an incorporation of imperfect interfaces with either LC- or HC-type interfaces, thermal invisibility conditions can still be established exactly.

In this study, we consider the configurations of a composite cylinder or sphere, with an introduction of a core material. Our motivations are that, on one aspect, one can adjust the material parameter of the core conductivity to fulfill the thermal transparency condition. On the other hand, we will be able to tune up the thermal intensity field within our targeted region to enhance or to shield the thermal intensity for different choices of core conductivity. We will demonstrate that the core material is significant in the design of metamaterial.

To begin with, we first introduce the mathematical framework of imperfect interfaces, or referred to as non-ideal interfaces. Two different types of imperfect interfaces are considered, low-conductivity (LC-) type and high-conductivity (HC-) type. LC-type interface is exactly the well-known Kapitza's thermal resistance [34]. The mathematical framework of LC-type interface can be derived by considering a thin interphase layer, of uniform thickness t with relatively low conductivity k_c [40], with an interface parameter characterized by $\beta = \lim_{t \rightarrow 0, k_c \rightarrow 0} k_c/t$, based on an asymptotic analysis. For an LC-type interface, the temperature field will have a drop across the interface, while the normal component of heat flux $\mathbf{q} \cdot \mathbf{n} = -(\mathbf{k} \nabla T) \cdot \mathbf{n}$ will be continuous

$$k_1 \frac{\partial T_1}{\partial r} = k_r \frac{\partial T_2}{\partial r} = \beta(T_2 - T_1)|_{r=a}, \quad (5)$$

here the interface parameter β , with a dimension of $\text{W} \cdot \text{m}^{-2} \text{K}^{-1}$, is defined as the ratio between the temperature drop and the heat flow across the interface. The case of $\beta \rightarrow \infty$ will represent perfect bonding interface, while $\beta \rightarrow 0$ stands for adiabatic (vacuous) contact [41]. Next, we consider high-conductivity (HC-) type of interface. HC-type interface [42] is physically opposite to that of LC-type interface. A mathematical theory of HC-type interface can be simulated by considering a thin interphase layer of a constant thickness t with relatively high conductivity k_c , with an interface parameter $\alpha = \lim_{t \rightarrow 0, k_c \rightarrow \infty} k_c t$. Here α has a

dimension of $\text{W} \cdot \text{K}^{-1}$. Perfect interfaces will correspond to $\alpha \rightarrow 0$, whereas $\alpha \rightarrow \infty$ describes contact with a medium of infinite conductivity [41]. We mention that in the context of elasticity HC-type interfaces are analogous to interface conditions with surface stress effects [43]. Specifically, for the HC-type interface the temperature field is continuous across the interface, while the jump of the normal component of heat flux is proportional to the surface Laplacian of the temperature field at the contact boundary. The interface conditions are

$$T_1 = T_2|_{r=a}, \quad \left(k_1 \frac{\partial T_1}{\partial r} - k_r \frac{\partial T_2}{\partial r} \right) \Big|_{r=a} = (\alpha \Delta_s T_1)|_{r=a}, \quad (6)$$

where Δ_s is the surface Laplacian, expressed as $\Delta_s = (1/r^2)(\partial^2/\partial\theta^2)$ for cylindrical coordinates, and as

$$\Delta_s = \frac{1}{r^2 \sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial}{\partial\theta} \right) + \frac{1}{r^2 \sin^2\theta} \frac{\partial^2}{\partial\varphi^2}, \quad (7)$$

for spherical coordinates [44]. Note that here θ is the cone angle measured from the z axis (latitudinal angle) ($0 \leq \theta \leq \pi$) and φ is the

polar angle from the x - y plane (azimuthal angle) ($0 \leq \varphi \leq 2\pi$). In our three-dimensional configuration, owing to symmetry, the temperature field will be independent of φ . In the contexts of composite materials, the effect of LC- and HC-type interface have been studied extensively [40,45–47]. We mention that in addition to LC- and HC-type interfaces, there is a third model in which both the temperature and the normal heat flux could have possible jumps across the interface. This model is to simulate a three-phase medium as a two-phase medium but with a general imperfect interface condition [48–50].

We first outline our main results and defer the description of analytical methods and derivations in Section 3 and Appendix A. For convenience of expositions of the exact solutions, we define some abbreviations of symbols in Table 1 for circular and spherical configurations.

2.1. Composite cylinder configuration

The thermal transparency condition can be established from a homogenization procedure to be outlined in Section 3. For composite cylinder systems, the thermal transparency condition can be derived as

$$k_0 = k_G \frac{(\tilde{g} + 1) + c^\lambda(\tilde{g} - 1)}{(\tilde{g} + 1) - c^\lambda(\tilde{g} - 1)}, \quad (8)$$

for an LC-type interface, and

$$k_0 = k_G \frac{(\hat{g} + 1) + c^\lambda(\hat{g} - 1)}{(\hat{g} + 1) - c^\lambda(\hat{g} - 1)}, \quad (9)$$

for an HC-type interface. When the interface is perfectly bonded, by letting $\beta \rightarrow \infty$ in (8) or $\alpha = 0$ in (9), we can deduce the thermal transparency condition for composite cylinder with ideal interface as [51]

$$k_0 = k_G \frac{(g + 1) + c^\lambda(g - 1)}{(g + 1) - c^\lambda(g - 1)}. \quad (10)$$

To establish more clearly the analogy between k_1 and $a\beta$, we normalize the core conductivity k_1 and the interface parameter β with respect to the value of k_G . It turns out that the expression for k_0/k_G has a symmetric pattern

$$\frac{k_0}{k_G} = \frac{\left(1 + \frac{k_G}{k_1} + \frac{k_G}{a\beta}\right) + c^\lambda \left(1 - \frac{k_G}{k_1} - \frac{k_G}{a\beta}\right)}{\left(1 + \frac{k_G}{k_1} + \frac{k_G}{a\beta}\right) - c^\lambda \left(1 - \frac{k_G}{k_1} - \frac{k_G}{a\beta}\right)}, \quad (11)$$

Table 1

Definition of relevant quantities of thermal transparency conditions for circular and spherical configurations with imperfect interface.

	Circular configuration (2D)	Spherical configuration (3D)
volume fraction	$c = (a/b)^2$	$c = (a/b)^3$
anisotropy parameter	$\lambda = \sqrt{k_\theta/k_r}$	$\lambda = \left(\sqrt{1 + 8k_t/k_r} - 1\right)/2$
Equivalent conductivity of Region II	$k_G = \lambda k_r = \sqrt{k_r k_\theta}$	$k_G = \lambda k_r = \frac{2k_t}{\lambda + 1}$
Conductivity ratio LC-type interface	$g = k_1/k_G$ $\tilde{k}_1 = \left(k_1^{-1} + (a\beta)^{-1}\right)^{-1} = \frac{k_1}{1 + \tilde{\beta}}$ where $\tilde{\beta} = k_1/a\beta$	$g = k_1/k_G$ $\tilde{k}_1 = \left(k_1^{-1} + (a\beta)^{-1}\right)^{-1} = \frac{k_1}{1 + \tilde{\beta}}$ where $\tilde{\beta} = k_1/a\beta$
HC-type interface	$\hat{k}_1 = k_1 + \frac{\alpha}{a} = k_1(1 + \hat{\alpha})$ where $\hat{\alpha} = \alpha/ak_1$	$\hat{k}_1 = k_1 + \frac{2\alpha}{a} = k_1(1 + 2\hat{\alpha})$ where $\hat{\alpha} = \alpha/ak_1$
LC-type modified conductivity ratio	$\tilde{g} = \tilde{k}_1/k_G = \frac{g}{1 + \tilde{\beta}}$	$\tilde{g} = \tilde{k}_1/k_G = \frac{g}{1 + \tilde{\beta}}$
HC-type modified conductivity ratio	$\hat{g} = \hat{k}_1/k_G = g(1 + \hat{\alpha})$	$\hat{g} = \hat{k}_1/k_G = g(1 + 2\hat{\alpha})$

for configurations with LC-interface, and has the form

$$\frac{k_0}{k_G} = \frac{\left(1 + \frac{k_1}{k_G} + \frac{\alpha}{ak_G}\right) - c^\lambda \left(1 - \frac{k_1}{k_G} - \frac{\alpha}{ak_G}\right)}{\left(1 + \frac{k_1}{k_G} + \frac{\alpha}{ak_G}\right) + c^\lambda \left(1 - \frac{k_1}{k_G} - \frac{\alpha}{ak_G}\right)}, \quad (12)$$

for configurations with HC-interface.

The expressions in (11) and (12) have an interesting interpretation. We find that the pair quantities are entirely equivalent

$$\frac{k_G}{k_1} \leftrightarrow \frac{k_G}{a\beta}, \quad \frac{k_1}{k_G} \leftrightarrow \frac{\alpha}{ak_G}, \quad (13)$$

for LC- and HC- type interfaces, respectively. That is, the effect of core conductivity k_1 on the effect of the background conductivity k_0 is the same with that of $a\beta$. In fact, the value of thermal transparency condition k_0/k_G will only be affected by the sum of $k_G/k_1 + k_G/a\beta$. Likewise, the value k_0/k_G for a system with HC-type interface will only be affected by the sum of $k_1/k_G + \alpha/ak_G$. Also, it can be easily identified that a dual relationship [52–54] exists for the mathematical framework of two-dimensional conductivity.

For an illustration, we present a three-dimensional plot in Fig. 2 the value of k_0/k_G versus the quantities of k_1/k_G and $a\beta/k_G$. Note that the 3D plot possesses a symmetry pattern along the axes of k_1/k_G and $a\beta/k_G$. Each color corresponds to different values of anisotropic parameter λ . To perceive the value of k_0/k_G versus the effect of imperfect parameters for different values of anisotropic parameters λ , we present in Fig. 3 for different values of k_1/k_G . In each figure, the left panel indicates the

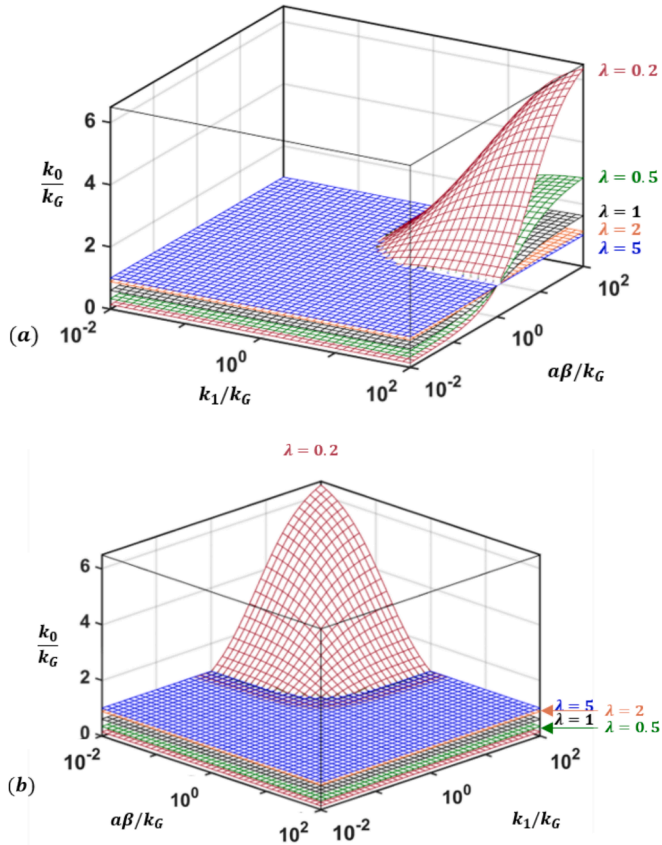


Fig. 2. A 3D-plot of the value of k_0/k_G for a two-dimensional configuration versus the parameters of k_1/k_G and $a\beta/k_G$. Each color corresponds to different value of anisotropic parameters λ . The area fraction c for numerical illustration is specified as $c = 1/4$. Note that the 3D plot possesses a symmetry pattern along the axes of k_1/k_G and $a\beta/k_G$. (a) A 3D plot from 30 degree clockwise. (b) A 3D plot from a different angle, -45 degree counterclockwise.

effects of LC-type interface and the right panel represents the effects of HC-type interface. It turns out that a smaller value of λ , that is $k_r > k_\theta$, will have a pronounced effect on the value of k_0/k_G . On the other hand, a larger value of λ has little effect on k_0/k_G . It is seen that the middle dashed line, corresponding to $\beta \rightarrow \infty$ or $\alpha \rightarrow 0$, is the situation of perfect interface. The very left end in Fig. 3 represents adiabatic (vacuous) contact $\beta \rightarrow 0$, which is the situation of least conduction. While on the very right end $\alpha \rightarrow \infty$ it describes a contact with infinite conductivity.

Next, we present the thermal intensity within our targeted region, namely Region I, versus the uniform applied intensity along the x direction for a 2D configuration. We mention that the temperature fields in Region I and Region III are linear functions of x , and thus quantitatively the ratio of thermal intensity is exactly the temperature ratio in these two regions. For clarity, the uniform intensity is our targeted region is denoted by E_1 and the uniform prescribed intensity in the x -direction is denoted by E . We note that under thermal transparency the thermal intensity in region III, E_3 , will be the same as E . We are particularly concerned with extreme values of thermal intensity that could possibly take on a maximum or minimum value. Specifically, a value of E_1/E_3 represents a concentrating effect, while a value of $E_1/E_3 < 1$ indicates a shielding effect. The local field are resolved from the governing field equations and will be described later in Section 3 and in Appendix A. Specifically, the ratio of thermal intensity in Region I versus the uniform applied intensity is

$$\frac{E_1}{E_3} \Big|_\beta = \frac{c^{-\frac{1}{2}}}{\left(1 + \tilde{\beta}\right) \cosh\left(\frac{\lambda}{2} \ln c\right) \left[1 - \frac{g}{1+\beta} \tanh\left(\frac{\lambda}{2} \ln c\right)\right]}, \quad (14)$$

for a configuration with LC-type interface, and has the form

$$\frac{E_1}{E_3} \Big|_\alpha = \frac{c^{-\frac{1}{2}}}{\cosh\left(\frac{\lambda}{2} \ln c\right) \left[1 - g(1 + \hat{\alpha}) \tanh\left(\frac{\lambda}{2} \ln c\right)\right]}, \quad (15)$$

for a configuration with HC-type interface. Note that for perfect bonding interfaces we can set $\beta \rightarrow \infty$ or $\alpha \rightarrow 0$ in (14) and (15), respectively, and deduce the exact solution

$$\frac{E_1}{E_3} \Big|_p = \frac{c^{-\frac{1}{2}}}{\cosh\left(\frac{\lambda}{2} \ln c\right) \left[1 - \frac{k_1}{k_G} \tanh\left(\frac{\lambda}{2} \ln c\right)\right]}. \quad (16)$$

To illustrate the thermal intensity ratio within the core region versus the uniform applied intensity, we present in Fig. 4 the value of E_1/E_3 versus the interface parameters $a\beta/k_G$ and α/ak_G . It is seen that the maximum intensity will occur for systems with perfect interface and that a higher intensity ratio will appear for systems with a smaller λ . Also, a smaller value of k_1/k_G will lead to a higher intensity ratio in Region I.

2.2. Composite sphere configuration

Next, we consider a three-dimensional composite sphere configuration (Fig. 1b). The thermal transparency connection will be constructed via a two-step homogenization procedure described in Section 3. For composite sphere configurations, the thermal transparency condition can be found as

$$k_0 = k_G \frac{\left(\tilde{g} + \frac{\lambda+1}{\lambda}\right) + c^{\frac{1+2\lambda}{3}} \frac{\lambda+1}{\lambda} (\tilde{g} - 1)}{\left(\tilde{g} + \frac{\lambda+1}{\lambda}\right) - c^{\frac{1+2\lambda}{3}} (\tilde{g} - 1)}, \quad (17)$$

for systems with LC-type interface, and has the form

$$k_0 = k_G \frac{\left(\hat{g} + \frac{\lambda+1}{\lambda}\right) + c^{\frac{1+2\lambda}{3}} \frac{\lambda+1}{\lambda} (\hat{g} - 1)}{\left(\hat{g} + \frac{\lambda+1}{\lambda}\right) - c^{\frac{1+2\lambda}{3}} (\hat{g} - 1)}, \quad (18)$$

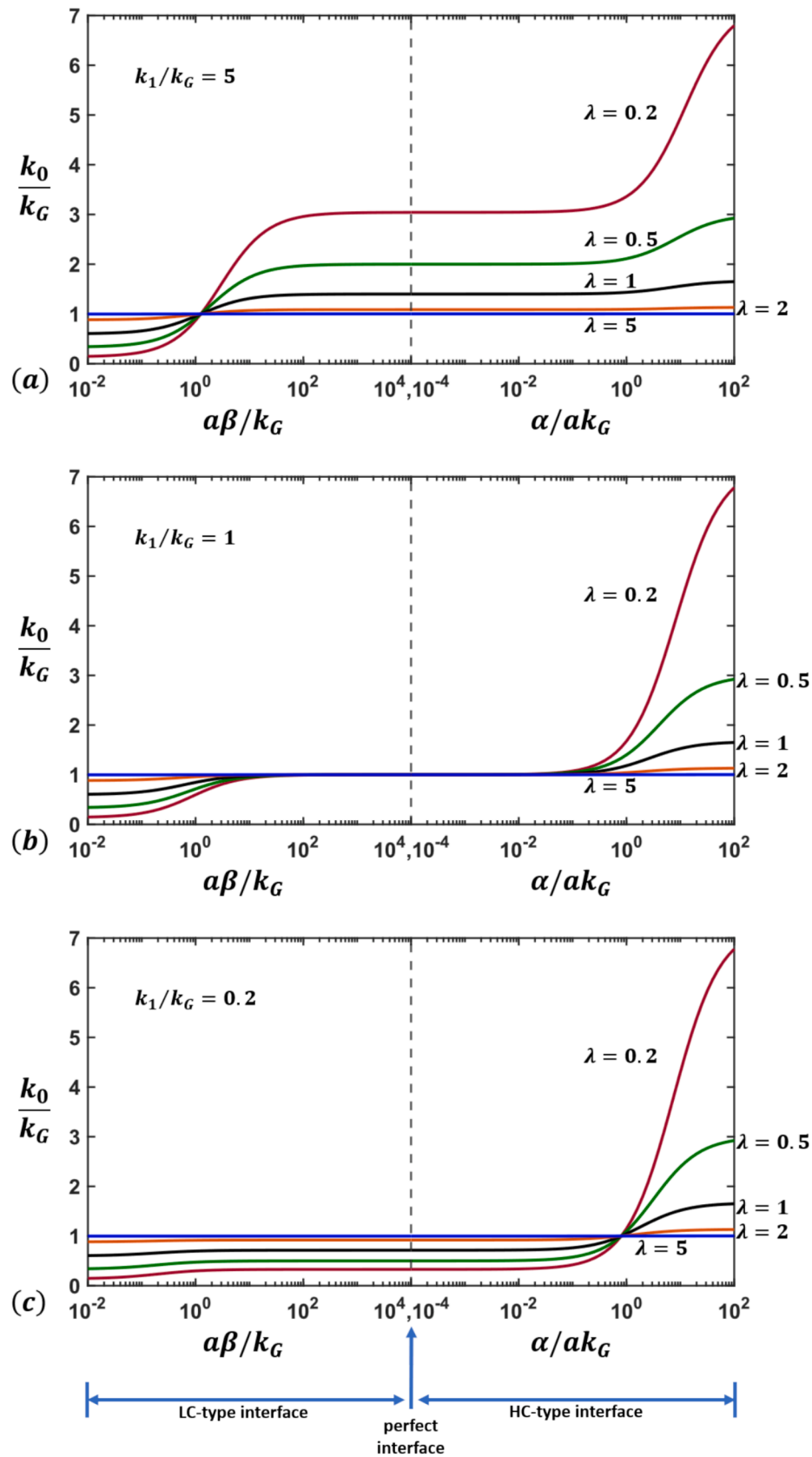


Fig. 3. A line chart of k_0/k_G for composite cylinder configuration versus the normalized interface parameters, where the left panel is for the LC-type interface and the right panel is the HC-type interface. The dashed line in the middle represents the situation of perfect interface. Different colors represent different anisotropic parameter λ . Fig. 3(a)-3(c) correspond respectively to $k_1/k_G = 5, 1, 0.2$. The area fraction c for numerical illustration is specified as $c = 1/4$.

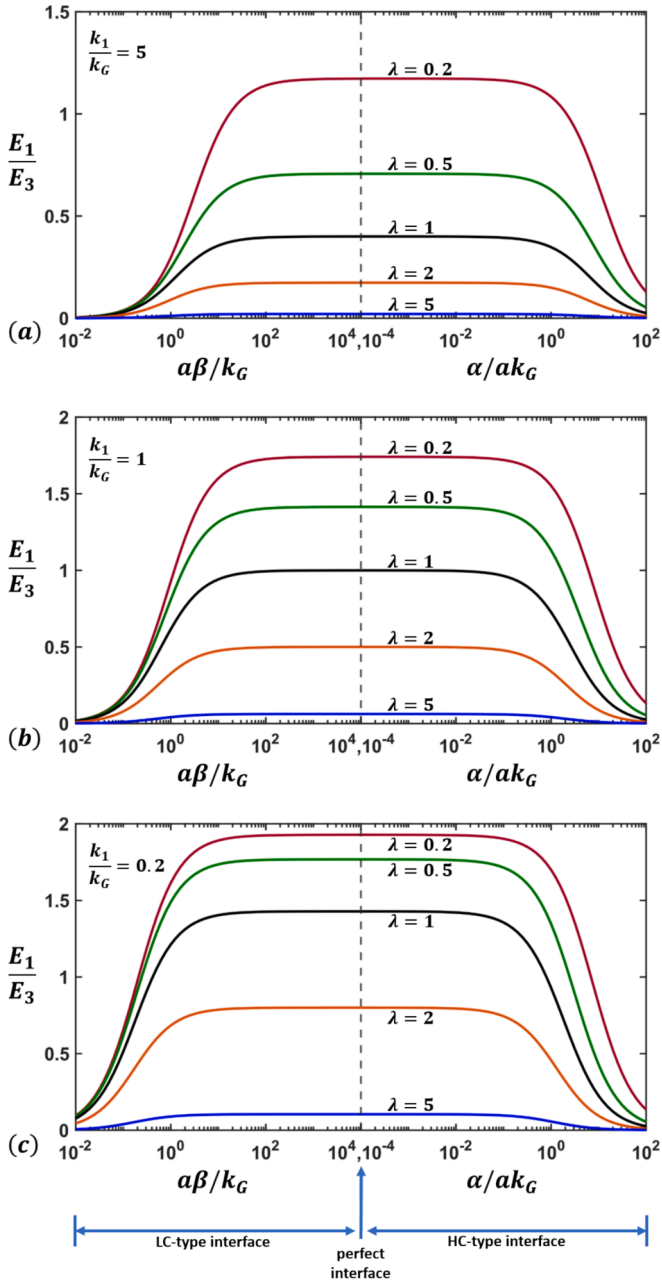


Fig. 4. An illustration of the value of E_1/E_3 versus the parameters of $a\beta/k_G$ and α/ak_G . Each color corresponds to different value of anisotropic parameters λ . The area fraction c for numerical illustration is specified as $c = 1/4$. Fig. 4(a)–4(c) correspond respectively to $k_1/k_G = 5, 1, 0.2$.

for systems with HC-type interface. It is seen that the expressions (17) and (18) are exactly identical except that $\tilde{g}$ in (17) is replaced by $\hat{g}$ in (18).

Interestingly, when the core conductivity k_1 and the interface parameter β are normalized with respect to the value of k_G , the expression k_0/k_G in (17) for LC-type interface can be written as

$$\frac{k_0}{k_G} = \frac{\left[1 + \frac{\lambda+1}{\lambda} \left(\frac{k_G}{k_1} + \frac{k_G}{a\beta}\right)\right] + c^{\frac{1+2\lambda}{3}} \frac{\lambda+1}{\lambda} \left(1 - \frac{k_G}{k_1} - \frac{k_G}{a\beta}\right)}{\left[1 + \frac{\lambda+1}{\lambda} \left(\frac{k_G}{k_1} + \frac{k_G}{a\beta}\right)\right] - c^{\frac{1+2\lambda}{3}} \frac{\lambda+1}{\lambda} \left(1 - \frac{k_G}{k_1} - \frac{k_G}{a\beta}\right)}, \quad (19)$$

and the expression in (18) for HC-type interface will become

$$\frac{k_0}{k_G} = \frac{\left(\frac{k_1}{k_G} + \frac{2a}{ak_G} + \frac{\lambda+1}{\lambda}\right) + c^{\frac{1+2\lambda}{3}} \frac{\lambda+1}{\lambda} \left(\frac{k_1}{k_G} + \frac{2a}{ak_G} - 1\right)}{\left(\frac{k_1}{k_G} + \frac{2a}{ak_G} + \frac{\lambda+1}{\lambda}\right) - c^{\frac{1+2\lambda}{3}} \frac{\lambda+1}{\lambda} \left(\frac{k_1}{k_G} + \frac{2a}{ak_G} - 1\right)}. \quad (20)$$

Again, as in two-dimensional configurations with LC-type interface, this suggests that the value of k_0/k_G in (19) for a three-dimensional composite sphere configuration depends on the the sum of $k_G/k_1 + k_G/a\beta$, while for a three-dimensional configuration with HC-type interfaces Eq. (20) will now be affected by the sum $k_1/k_G + 2a/ak_G$. In contrast to (13), the effect of k_1/k_G is now relevant to the quantity $2a/ak_G$ for a three-dimensional system. For the situations of perfect bonding interface, we can let $\beta \rightarrow \infty$ or $\alpha = 0$ in (17) and (18), and reduce the exact connection

$$\frac{k_0}{k_G} = \frac{\left(\frac{k_1}{k_G} + \frac{\lambda+1}{\lambda}\right) + c^{\frac{1+2\lambda}{3}} \frac{\lambda+1}{\lambda} \left(\frac{k_1}{k_G} - 1\right)}{\left(\frac{k_1}{k_G} + \frac{\lambda+1}{\lambda}\right) - c^{\frac{1+2\lambda}{3}} \frac{\lambda+1}{\lambda} \left(\frac{k_1}{k_G} - 1\right)}. \quad (21)$$

When the anisotropic shell is further simplified as an isotropic layer with $\lambda = 1$, the expression (21) will recover the well-known effective conductivity of composite sphere assemblages [55].

To perceive the thermal transparency condition for a composite sphere system, we present in Fig. 5 the value of k_0/k_G versus the effect of imperfect parameter for different values of anisotropic parameters λ . Again, as in Fig. 3 for composite cylinder systems, the left panel is the result for the effects of LC-type interface, while the right panel represents the effects of HC-type interface. The overall trends for λ in Fig. 5 for composite sphere systems and in Fig. 3 for composite cylinder systems are similar, but the value of k_0/k_G for three-dimensional configurations will be higher.

Next, we present the thermal intensity field in the core region for a composite sphere system with imperfect interface. The local field will be directly solved from the field equations to be outlined later in Section 3 and Appendix A. Specifically, the results are

$$\frac{E_1}{E_3}\bigg|_{\beta} = \frac{k_0}{k_1} \frac{c^{-\frac{1}{2}}}{\cosh\left(\frac{1+2\lambda}{6} \text{Inc}\right) \left\{1 - \frac{1}{1+2\lambda} \left[\frac{2(1+\lambda)}{g} - 1\right] \tanh\left(\frac{1+2\lambda}{6} \text{Inc}\right)\right\}}, \quad (22)$$

for LC-type interface, and

$$\frac{E_1}{E_3}\bigg|_{\alpha} = \frac{c^{-\frac{1}{2}}}{\cosh\left(\frac{1+2\lambda}{6} \text{Inc}\right) \left\{1 - \frac{2\lambda g+1}{1+2\lambda} \tanh\left(\frac{1+2\lambda}{6} \text{Inc}\right)\right\}}, \quad (23)$$

for HC-type interface.

We note that, when $k_1 = k_0$, Eq.(22) will recover the solution of previous result in which the composite sphere is a hollow spherical shell [38, Eq.7 and Eq.9₁]. On the other hand, we note that Eq.(23) is exactly the same expression as that of Chen and Lin [38, Eq.7 and Eq.9₂]. For systems with perfect bonding interface, we can deduce from either (22) or (23) as

$$\frac{E_1}{E_3}\bigg|_p = \frac{c^{-\frac{1}{2}}}{\cosh\left(\frac{1+2\lambda}{6} \text{Inc}\right) \left[1 - \frac{2\lambda g+1}{1+2\lambda} \tanh\left(\frac{1+2\lambda}{6} \text{Inc}\right)\right]}. \quad (24)$$

To illustrate the thermal intensity ratio within the inclusion versus the uniform applied intensity, we present in Fig. 6 the value of E_1/E_3 versus the interface parameters $a\beta/k_G$ and α/ak_G . As in composite cylinder systems, the maximum intensity will occur for systems with perfect interface and that a higher intensity will appear in systems with a smaller value of λ . In addition, to attain a thermal concentrating effect, the core material must be less conductive. A thermal cloaking effect with low intensity ratio within the core region will occur for either $a\beta/k_G \rightarrow 0$ or $\alpha/ak_G \rightarrow \infty$. This illustration demonstrates that the thermal

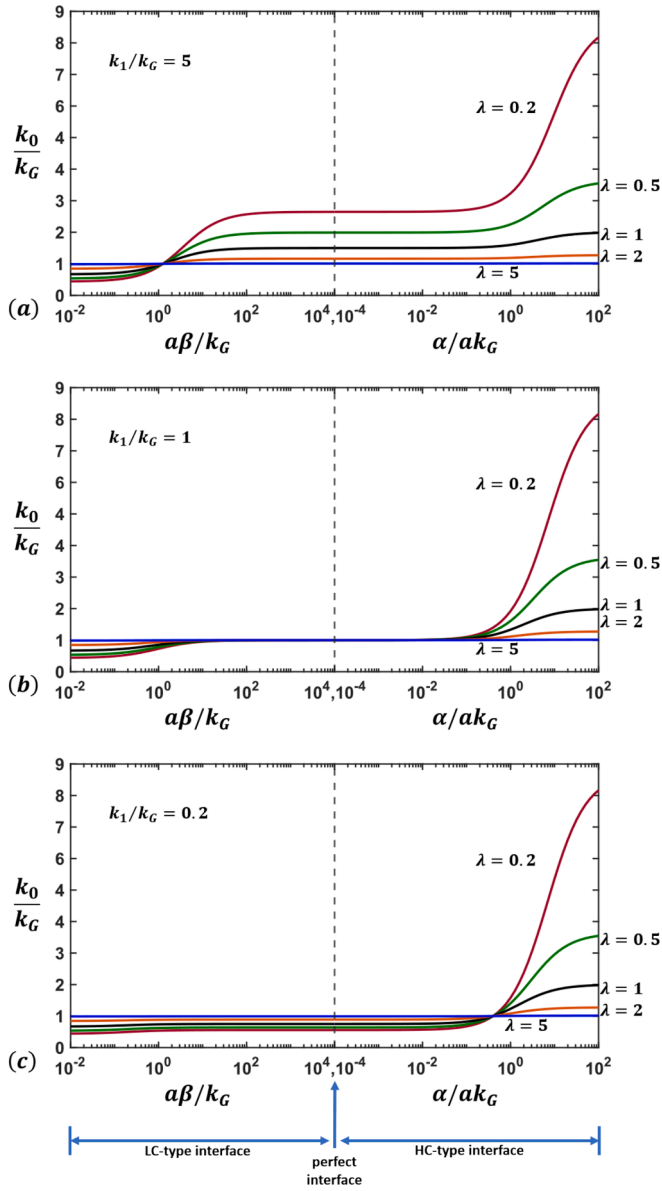


Fig. 5. A line chart of k_0/k_G for the configuration of composite sphere versus the normalized interface parameters, where the left panel is for the LC-type interface and the right panel is the HC-type interface. The dashed line in the middle represents the perfect interface. The volume fraction c for numerical illustration is specified as $c = 1/4$. Different colors represent different anisotropic parameter λ . Fig. 5(a)-3(c) correspond respectively to $k_1/k_G = 5, 1, 0.2$.

intensity within Region I can be completely shielded for systems containing imperfect interface with an extreme interface parameter, while the thermal intensity within Region I can be enhanced but will be bounded by a geometric ratio b/a . Also, to illustrate the value of E_1/E_3 versus the effect of inclusion conductivity k_1/k_G , we present in Fig. 7(a) and Fig. 7(b) for composite cylinder and sphere systems with perfect interface. This illustration is that the intensity ratio will give a maximum value for perfect bonding interface. Again, we highlight different values of anisotropic parameters λ with different colors, in which a smaller value of λ will give a higher intensity ratio. For the effect of core conductivity, we note that, for both two and three-dimensional systems, a higher intensity ratio will occur when the core has a lower conductivity.

Lastly, we examine the influence of volume fraction with perfect interface on the effect of thermal intensity within Region I. For $c \rightarrow 1$, it can be readily seen from the analytical expressions, (14), (15), (22), and (23), that the ratio of thermal intensity ratio $E_1/E_3 \rightarrow 1$. For $c \rightarrow 0$, the core

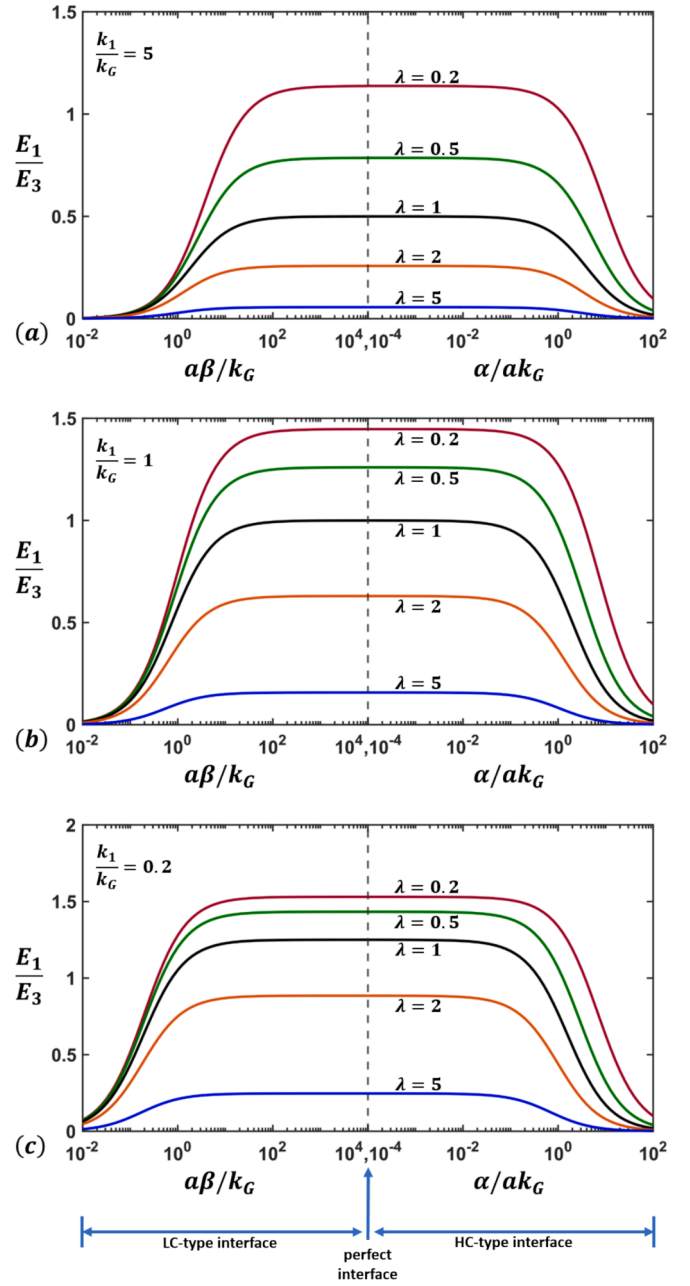


Fig. 6. An illustration of the value of E_1/E_3 for three-dimensional configurations versus the parameters of $a\beta/k_G$ and α/ak_G . Each color corresponds to different value of anisotropic parameters λ . The volume fraction c for numerical illustrations is specified as $c = 1/4$. Fig. 6(a)-(c) correspond respectively to $k_1/k_G = 5, 1, 0.2$.

region will nearly vanish. We present in Fig. 8 numerical illustrations of E_1/E_3 for composite cylinder and composite sphere configurations. As a smaller value of k_1/k_G will give a greater intensity ratio, we will only present the situation of $k_1/k_G = 0.2$ for brevity. We note, however, that the case of $c \rightarrow 0$, is close to the situation that the whole composite cylinder or sphere is constituted by a core region with curvilinearly anisotropic material. In the context of elasticity, previous studies have shown that a singular behavior could occur at the central point of a curvilinearly anisotropic core [56].

3. Analytical methods

Thermal transparency condition. The thermal transparency

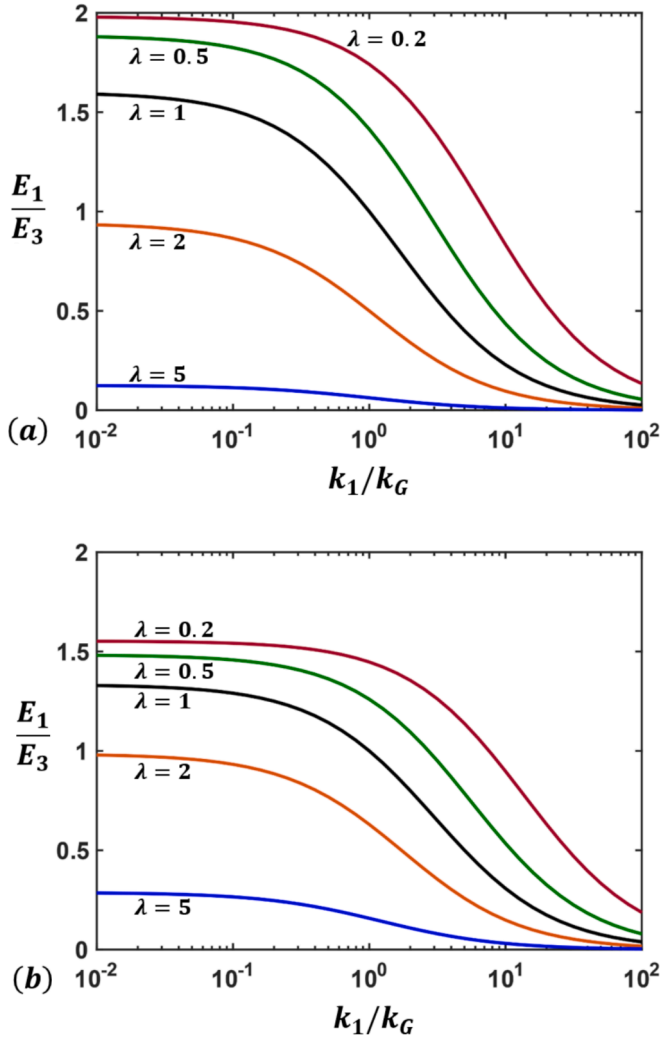


Fig. 7. An illustration of the value of E_1/E_3 of the composite systems with perfect interface versus the parameter of k_1/k_G . Each color corresponds to different value of anisotropic parameters λ . The volume fraction or area fraction c for numerical illustration is specified as $c = 1/4$. Fig. 7(a) corresponds to composite cylinder configurations, while Fig. 7(b) is for composite sphere configurations.

condition for composite cylinder configurations were given in Eq. (8) for LC-type interface and in Eq. (9) for HC-type interface. These exact conditions can be constructed based on the concept of neutral inclusions [35] via a series of homogenization procedures. To construct the thermal transparency conditions, we utilize the steps [52, Fig. 6] based on the theory of neutral inclusion. In the present configuration of composite cylinder, the procedures can be separated into two steps. The first step is to consider core region, Region I, with the effect of imperfect interface at $r = a$. This will lead to an equivalent circular inclusion with an effective isotropic conductivity $\tilde{k}_1$ or $\hat{k}_1$. Secondly, a composite cylinder model, with an isotropic inclusion with conductivity $\tilde{k}_1$ or $\hat{k}_1$ coated with a cylindrically anisotropic coating layer, is determined. This will give an equivalent conductivity perfectly matched to the background medium. The coated circle will become thermally transparent if the effective conductivity of the composite cylinder is identical with that of the background material. Similarly, for three-dimensional configurations, thermal transparent conditions with either LC-type or HC-type interface can be derived in similar ways [38, Fig. 5] based on composite sphere assemblages.

Thermal intensity in the core region. We are concerned with the

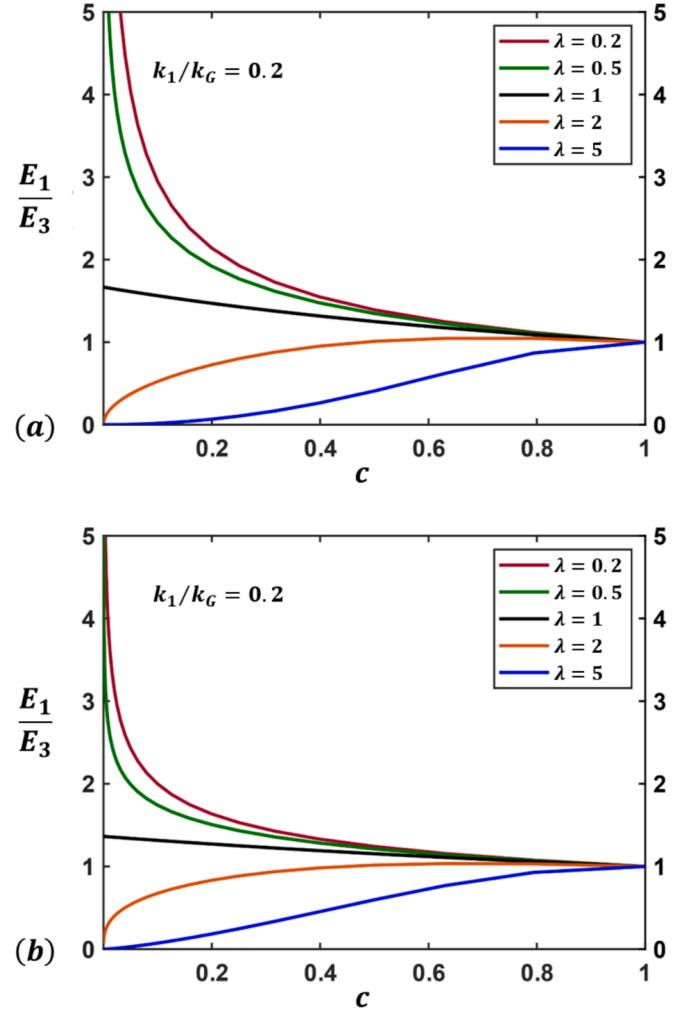


Fig. 8. An illustration of the value of E_1/E_3 of the composite systems with perfect interface versus the parameter of c . Fig. 8(a) correspond to composite cylinder configurations, while Fig. 8(b) is for composite sphere configurations. Each color corresponds to different values of anisotropic parameter λ .

thermal intensity within the core region versus the uniform applied intensity. Here the intensity field needs to be solved from the boundary value problem. The formulation is briefly outlined here and detailed results are given in Appendix A. Under a steady-state condition, in the absence of heat source, the temperature fields in Regions I and III are governed simply by Laplace equation. For Region II, the field equation with the conductivity tensor (1) can be written as $\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} + \frac{\lambda^2}{r^2} \frac{\partial^2 T}{\partial \theta^2} = 0$ in a two-dimensional configuration, while the field equation is $r^2 \frac{\partial^2 T}{\partial r^2} + 2r \frac{\partial T}{\partial r} + \frac{k_r}{k_t} \left(\frac{\partial^2 T}{\partial \rho^2} + \cot \theta \frac{\partial T}{\partial \theta} \right) = 0$ for a three-dimensional configuration. Here λ was given in Table 1 for both configurations. For a two-dimensional system, we impose the boundary condition $T(x = r \cos \theta = \pm x_0) = \mp T_0$ on the left and right surfaces of the configuration (Fig. 1a), with $T_0 = -Ex_0$ and E is the prescribed constant thermal intensity. Taking into account the boundary condition and the geometric symmetry, the temperature fields can be expressed as [4]

$$\begin{aligned} T_1 &= A_1 r \cos \theta, r \leq a, \\ T_2 &= (A_2 r^\lambda + B_2 r^{-\lambda}) \cos \theta, a \leq r \leq b, \\ T_3 &= -E r \cos \theta, r > b, \end{aligned} \quad (25)$$

for the composite cylinder, and

$$\begin{aligned} T_1 &= A_1 r \cos \theta, r \leq a, \\ T_2 &= (A_2 r^\lambda + B_2 r^{-\lambda-1}) \cos \theta, a \leq r \leq b, \\ T_3 &= -E r \cos \theta, r > b, \end{aligned} \quad (26)$$

for the composite sphere. Note that, for a three-dimensional configuration, a uniform intensity is applied in the z -direction (Fig. 1b). Here the coefficients A_1, A_2, B_2 are constant coefficients to be determined from the boundary condition and the interface condition (5) for LC-type interface, or (6) for HC-type interface. Obviously, owing to the presence of Region I and II, the thermal intensity in the background medium, Region III, will be distorted from the prescribed intensity E . But now if we enforce that the outer field in Region III remains the same as the uniform prescribed intensity E , then the effect of composite cylinder or sphere, that is Region I and Region II altogether, can be viewed as thermally transparent. To determine the remaining coefficients, interface conditions need to be invoked. This will lead to four conditions to be determined from three coefficients together with the as yet unknown background material conductivity k_0 . This will provide 4 conditions to be satisfied by the three unknown coefficients A_1, A_2, B_2 , and an as-yet

undetermined background material conductivity k_0 . It appears that an exact expression for k_0 can be uniquely determined as in (8), (9), (17) and (18), referred to as thermally transparent conditions. Detailed procedures for the thermal transparency conditions and field solutions based on the neutral inclusion method will be given in Appendix A.

4. Validations with finite element simulations

In addition to analytical results, we also provide numerical simulations based on finite element calculations via the software COMSOL. To simulate the effect of imperfect interface, we model the imperfect interface as a thin layer of a constant thickness t with isotropic conductivity k_c , prescribed by either $k_1 t / a \tilde{\beta}$ (LC-type) or $\hat{a} k_1 / t$ (HC-type) [37,38]. Also, the curvilinearly anisotropic layer in Region II is modeled by a layered medium [57,58] with two alternating isotropic phases k_A and k_B , by letting $k_r = 2 / (k_A^{-1} + k_B^{-1})$, $k_\theta = (k_A + k_B) / 2$, based on a homogenization theory. For simplicity, we consider a specific example for

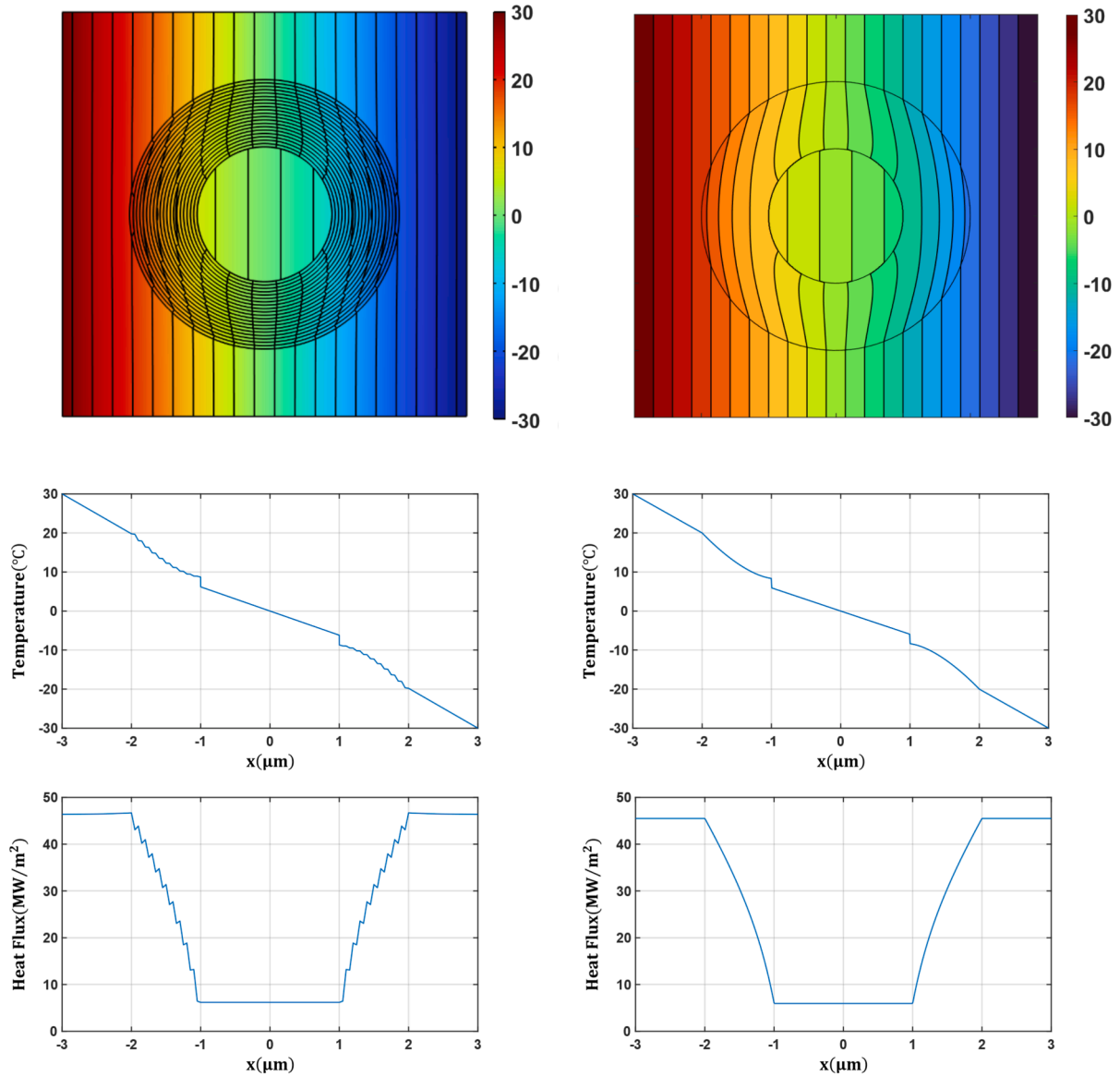


Fig. 9. Temperature and heat flux profiles corresponding to composite cylinder configurations with LC-type interface are shown based on analytical solutions and finite element calculations (COMSOL). The right panel is the analytical solutions for LC-type interfaces, while the left panel is the numerical simulations based on finite element calculations (COMSOL). In numerical calculations, we consider $a = 1 \mu\text{m}$, $b = 2 \mu\text{m}$ and $c = 1/4$. Also $k_1 = 1$, $k_r = 5/2$, $k_\theta = 10$, $k_A = 10 - 5\sqrt{3}$, $k_B = 10 + 5\sqrt{3}$, $\lambda = 2$, $k_G = 5$, $k_c = 0.00493$, $t = a/500$, $\beta = 2.465$, and $\tilde{\beta} = 0.40566$.

a composite cylinder configuration with two types of imperfect interface. Here in Fig. 9 we demonstrate the temperature and heat flux profiles in the x - y plane of composite cylinder systems with LC-type interface based on our analytical results and finite element simulations (COMSOL). In numerical calculations, we consider $a = 1 \mu\text{m}$, $b = 2 \mu\text{m}$, $c = 1/4$. Also, $k_1 = 1$, $k_r = 5/2$, $k_\theta = 10$, $k_A = 10 - 5\sqrt{3}$, $k_B = 10 + 5\sqrt{3}$, $\lambda = 2$, $k_G = 5$, $t = a/500$, $k_c = 0.00493$, $\beta = 2.465$, $\tilde{\beta} = 0.40566$. We model the curvilinearly anisotropic layer, Region II, by a 20-layer shell consisting of two alternating isotropic phases k_A and k_B . We mention that a 20-layer shell will give a converged solution with satisfactory accuracy in our calculations. The right panel is the analytical solutions for LC-type interfaces, while the left panel is the numerical simulations based on finite element calculations (COMSOL). To perceive the temperature drop or heat flux drop across the interface, a profile along the x -axis is highlighted just below the contour diagram. It is seen that both results agree well. In Fig. 10, we demonstrate the temperature and heat flux profiles of composite cylinder systems with HC-type interface based on our analytical results and finite element calculations (COMSOL). In numerical calculations, we consider $a = 1 \mu\text{m}$, $b = 2 \mu\text{m}$, $c = 1/4$. Also, $k_1 = 1$, $k_r = 5/2$, $k_\theta = 10$, $k_A = 10 - 5\sqrt{3}$, $k_B = 10 + 5\sqrt{3}$, $\lambda = 2$, $k_G = 5$, $t = a/500$, $k_c = 202.83$, $t = a/500$, $\alpha = 0.40566$, $\hat{\alpha} = 0.40566$. Again,

the curvilinearly anisotropic layer, Region II, is modeled by a 20-layer shell consisting of two alternating isotropic phases k_A and k_B . The right panel is the analytical solutions for HC-type interfaces, while the left panel is the numerical simulations based on finite element calculations (COMSOL). Again, it is seen that a 20-layer shell provides a smooth contour, and both results agree satisfactorily.

For a configuration of composite sphere system with imperfect interface, we consider a uniform intensity prescribed in the z -direction. We demonstrate in Fig. 11 the temperature and heat flux profiles of composite sphere systems with LC-type interface based on our analytical results and finite element simulations (COMSOL). In numerical calculations, we consider $a = 1 \mu\text{m}$, $b = 2 \mu\text{m}$, $c = 1/8$. Also, $k_1 = 1$, $k_r = 5/2$, $k_t = 15/2$, $k_A = 15 - (15 - 5\sqrt{6})/2$, $k_B = 15 - (15 + 5\sqrt{6})/2$, $\lambda = 2$, $k_G = 5$, $t = a/500$, $k_c = 0.00493$, $\beta = 2.465$, $\tilde{\beta} = 0.40566$. Again, the curvilinearly anisotropic layer, Region II, is modeled by a 20-layer shell consisting of two alternating isotropic phases k_A and k_B , which will give a converged solution with satisfactory accuracy. A temperature profile along the y - z plane is illustrated. The right panel is the analytical solutions of the contour of temperature for LC-type interfaces, while the left panel is the numerical simulations based on finite element calculations (COMSOL). To see the temperature drop or heat flux drop across the

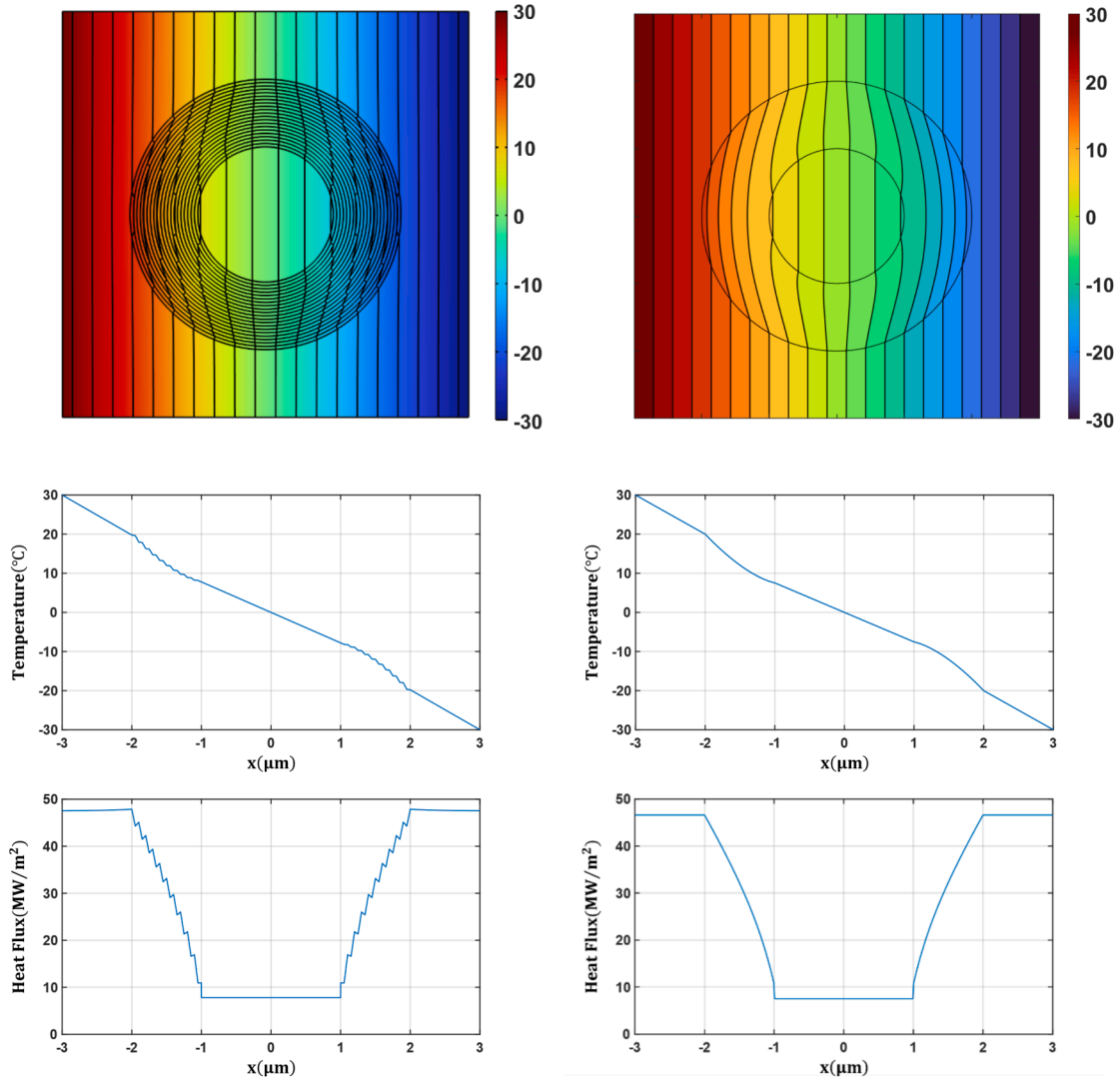


Fig. 10. Temperature and heat flux profiles corresponding to composite cylinder configurations with HC-type interface are shown based on analytical solutions and finite element calculations (COMSOL). The right panel is the analytical solutions for HC-type interfaces, while the left panel is the numerical simulations based on finite element calculations (COMSOL). In numerical calculations, we consider $a = 1 \mu\text{m}$, $b = 2 \mu\text{m}$ and $c = 1/4$. Also, $k_1 = 1$, $k_r = 5/2$, $k_\theta = 10$, $k_A = 10 - 5\sqrt{3}$, $k_B = 10 + 5\sqrt{3}$, $\lambda = 2$, $k_G = 5$, $k_c = 202.83$, $t = a/500$, $\alpha = 0.40566$, $\hat{\alpha} = 0.40566$.

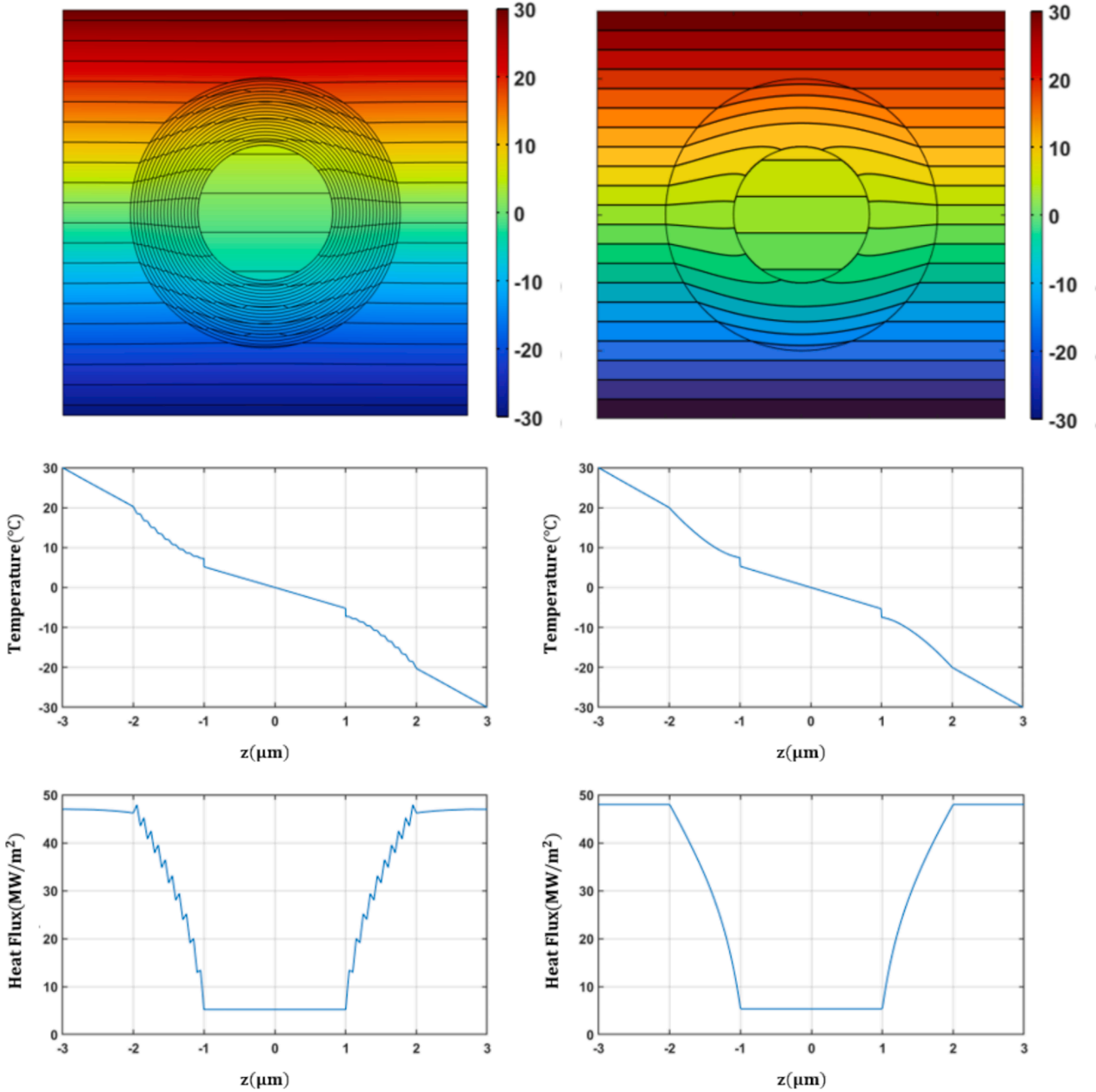


Fig. 11. Temperature and heat flux profiles along the y - z plane corresponding to composite sphere configurations with LC-type interface are shown based on analytical solutions and finite element calculations (COMSOL) under a uniform intensity along the z -axis. The right panel is the analytical solutions for LC-type interfaces, while the left panel is the numerical simulations based on finite element calculations (COMSOL). In numerical calculations, we consider $a = 1 \mu\text{m}$, $b = 2 \mu\text{m}$ and $c = 1/8$. Also $k_1 = 1$, $k_r = 5/2$, $k_t = 15/2$, $k_A = 15 - (15 - 5\sqrt{6})/2$, $k_B = 15 - (15 + 5\sqrt{6})/2$, $\lambda = 2$, $k_G = 5$, $k_c = 0.00493$, $t = a/500$, $\beta = 2.465$, and $\tilde{\beta} = 0.40566$.

interface, a profile along the z -axis is highlighted just below the contour diagram.

In Fig. 12, we demonstrate the temperature and heat flux profiles of composite sphere systems with HC-type interface. In numerical calculations, we consider $a = 1 \mu\text{m}$, $b = 2 \mu\text{m}$, $c = 1/8$. Also, $k_1 = 1$, $k_r = 5/2$, $k_t = 15/2$, $k_A = 15 - (15 - 5\sqrt{6})/2$, $k_B = 15 - (15 + 5\sqrt{6})/2$, $\lambda = 2$, $k_G = 5$, $t = a/500$, $k_c = 202.83$, $\alpha = 0.40566$, $\hat{\alpha} = 0.40566$. Again, the curvilinearly anisotropic layer is modeled by a 20-layer shell consisting of two alternating isotropic phases k_A and k_B . The right panel is the analytical solutions for HC-type interfaces, while the left panel is the numerical simulations based on finite element calculations (COMSOL). Again, both results agree satisfactorily.

5. Concluding Remarks

We show analytically that a simple configuration of composite cylinder or sphere can serve as a thermal device that is thermally transparent to the background medium, and has a tunable capability to enhance or to shield the thermal intensity within a targeted area. Effects of imperfect interface, with either low-conductivity or high-conductivity type, are incorporated exactly in the formulation. One of the keys in the thermal device is to introduce a homogeneous and curvilinearly anisotropic material as the coating layer to serve as a functional material to adjust the thermal intensity. In addition, we find out that the core material will play a crucial role in tuning the thermal transparency

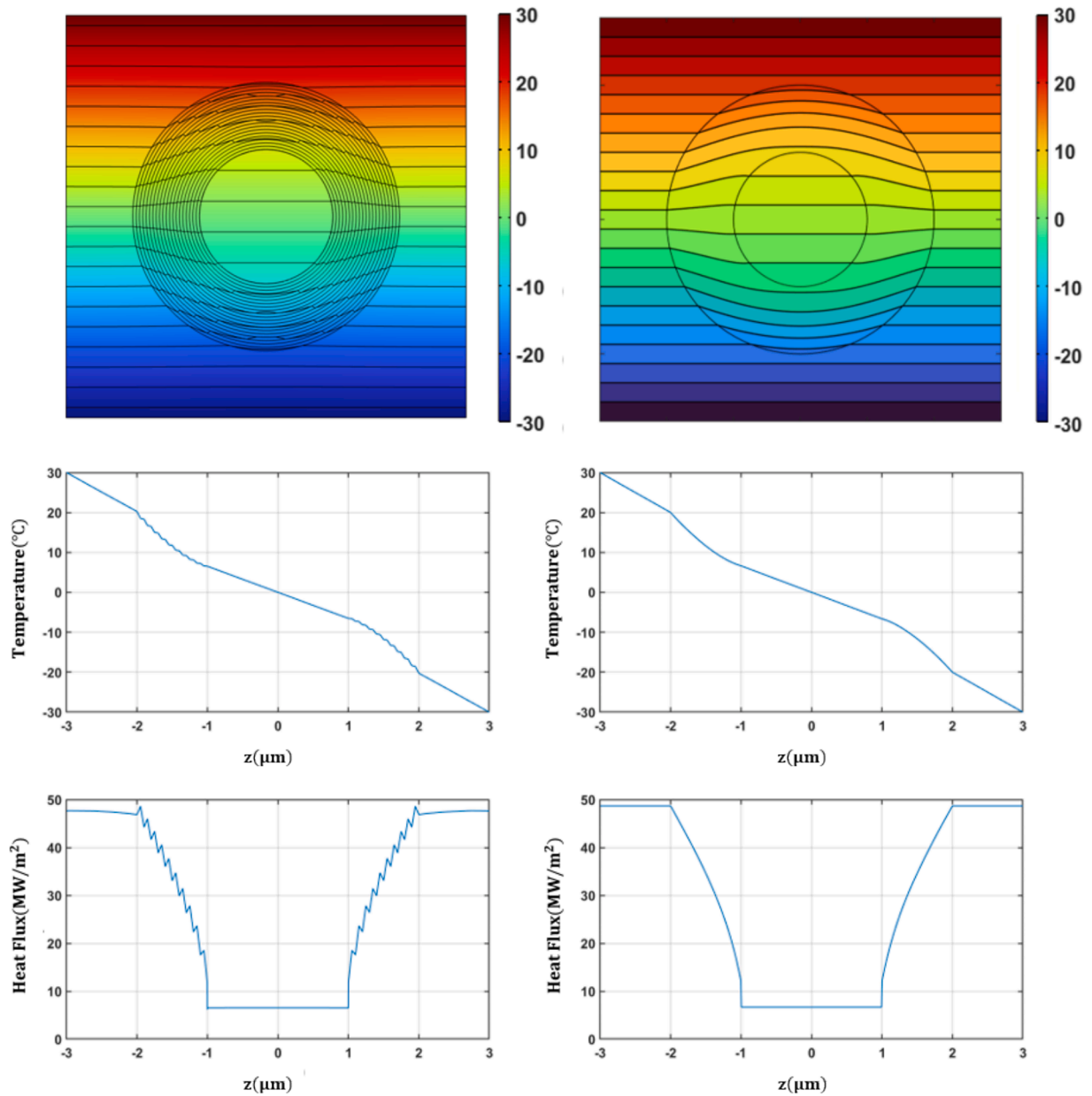


Fig. 12. Temperature and heat flux profiles along the y - z plane corresponding to composite sphere configurations with HC-type interface are shown based on analytical solutions and finite element calculations (COMSOL) under a uniform intensity along the z -axis. The right panel is the analytical solutions for HC-type interfaces, while the left panel is the numerical simulations based on finite element calculations (COMSOL). In numerical calculations, we consider $a = 1 \mu\text{m}$, $b = 2 \mu\text{m}$ and $c = 1/8$. Also $k_1 = 1$, $k_r = 5/2$, $k_t = 15/2$, $k_A = 15 - (15 - 5\sqrt{6})/2$, $k_B = 15 - (15 + 5\sqrt{6})/2$, $\lambda = 2$, $k_G = 5$, $k_c = 202.83$, $t = a/500$, $\alpha = 0.40566$, $\hat{\alpha} = 0.40566$.

connection and also to enhance the functional performance in the device. In addition, we find that, either in composite cylinder or in composite sphere configuration, the core conductivity can provide a thermal compensation quantitatively with the degree of imperfect bonding of either type. To be feasible in practice, we provide finite element simulations for systems in which the imperfect bonding interface is modeled as a thin interphase with either very low or very high conductivity, and the curvilinearly anisotropic layer is modeled as a layered medium with two different isotropic phases. The present study provides further insights to the understanding of heat transfer mechanisms across imperfect interfaces in the performances of simple composite cylinder or sphere configurations. This finding can serve as a general guideline in the design of composite structures as thermal metamaterials that facilitates broad applications.

CRediT authorship contribution statement

Ming-Cian Syu: Visualization, Validation, Software, Investigation, Formal analysis, Data curation. **Tungyang Chen:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

In this Appendix, we provide detailed derivation results for thermal transparency conditions and thermal field solutions summarized in Section 2, based on an effective medium theory of neutral inclusion. As depicted in Fig. 1, a uniform intensity E is prescribed in the x -direction for a 2D configuration (Fig. 1a), and along the z -direction for a 3D configuration (Fig. 1b). We are considering a composite cylinder or sphere, with possible imperfect interface of LC- or HC-type along the interface $r = a$, embedded within a homogenous medium so that the temperature field outside the composite cylinder or sphere will remain undisturbed, as if there is a homogeneous medium throughout. In this way, the composite cylinder or sphere is thermally transparent to the exterior background material, and the effective thermal conductivity of composite cylinder or sphere can be viewed as the same with that of the background medium, Region III. We first recall that the temperature fields in Region I, II and III were given in (25) for a composite cylinder configuration and in (26) for a composite sphere configuration, and the interface condition at $r = a$ was given in (5) for LC-type interface and in (6) for HC-type interface. The interface conditions at $r = a$ and $r = b$ will provide 4 conditions to be fulfilled from the unknown constants. We mention that as the composite cylinder or sphere is considered as a functional device, the interface between the composite structure and the outer background medium is taken as an perfect interface, that is

$$T_2 = T_3|_{r=b}, \quad k_2 \frac{\partial T_2}{\partial r} = k_0 \frac{\partial T_3}{\partial r} \Big|_{r=b}. \quad (A1)$$

For a composite cylinder configuration with LC-type interface, use of (25) in (5) and (A1) will provide 4 conditions

$$A_1 = \frac{k_r}{k_1} (\lambda A_2 a^{\lambda-1} - \lambda B_2 a^{-\lambda-1}), \quad (A2)$$

$$k_r (\lambda A_2 a^{\lambda-1} - \lambda B_2 a^{-\lambda-1}) = a\beta (A_2 a^{\lambda-1} + B_2 a^{-\lambda-1} - A_1), \quad (A3)$$

$$A_2 b^{\lambda-1} + B_2 b^{-\lambda-1} = -E, \quad (A4)$$

$$-k_0 E = k_r \lambda (A_2 b^{\lambda-1} - B_2 b^{-\lambda-1}), \quad (A5)$$

to be fulfilled from the 4 constants, A_1, A_2, B_2 and k_0 . Detailed solutions are

$$\begin{aligned} \tilde{A}_2 &= \frac{\left(g + 1 + \frac{k_1}{a\beta}\right)E}{c^\lambda \left(g - 1 - \frac{k_1}{a\beta}\right) - \left(g + 1 + \frac{k_1}{a\beta}\right)}, \\ \tilde{B}_2 &= \frac{-\left(g - 1 - \frac{k_1}{a\beta}\right)E}{c^\lambda \left(g - 1 - \frac{k_1}{a\beta}\right) - \left(g + 1 + \frac{k_1}{a\beta}\right)}, \end{aligned} \quad (A6)$$

where we have defined the notations $\tilde{A}_2 = A_2 a^{\lambda-1}$ and $\tilde{B}_2 = B_2 / a^{\lambda+1}$. The thermal transparency condition k_0 in (8) can then be derived from (A6) and (A5). Also, the temperature field, and the uniform intensity A_1 , in Region I in (14) can be obtained from (A6), (A2) and (25). For a composite cylinder configuration with HC-type interface, use of (25) in (6) and (A1) will provide 4 conditions

$$A_1 = A_2 a^{\lambda-1} + B_2 a^{-\lambda-1}, \quad (A7)$$

$$k_1 (A_2 a^{\lambda-1} + B_2 a^{-\lambda-1}) - k_G (A_2 a^{\lambda-1} - B_2 a^{-\lambda-1}) = -\frac{\alpha}{a} (A_2 a^{\lambda-1} + B_2 a^{-\lambda-1}), \quad (A8)$$

$$A_2 b^{\lambda-1} + B_2 b^{-\lambda-1} = -E, \quad (A9)$$

$$-k_0 E = k_r \lambda (A_2 b^{\lambda-1} - B_2 b^{-\lambda-1}). \quad (A10)$$

Explicit solutions of A_2 and B_2 are

$$\begin{aligned} \tilde{A}_2 &= \frac{\left(g + 1 + \frac{\alpha}{ak_G}\right)E}{c^\lambda \left(g - 1 + \frac{\alpha}{ak_G}\right) - \left(g + 1 + \frac{\alpha}{ak_G}\right)}, \\ \tilde{B}_2 &= \frac{-\left(g - 1 + \frac{\alpha}{ak_G}\right)E}{c^\lambda \left(g - 1 + \frac{\alpha}{ak_G}\right) - \left(g + 1 + \frac{\alpha}{ak_G}\right)}, \end{aligned} \quad (A11)$$

where $\tilde{A}_2$ and $\tilde{B}_2$ were defined as in (A6). Again thermal transparency condition k_0 in (9) can be derived from (A11) and (A10), and also the temperature field and also the uniform intensity in Region I in (15) can be found from (A11) and (A7).

For a composite sphere system, the derivation procedure is similar to that of composite cylinder system, but much complicated in algebra. Here now for systems with LC-type interface, use of (26) in (5) and (A1) will provide 4 conditions

$$A_1 = \frac{k_r}{k_1} (\lambda A_2 a^{\lambda-1} - (\lambda+1) B_2 a^{-\lambda-2}), \quad (A12)$$

$$k_G \left[A_2 a^{\lambda-1} - \left(1 + \frac{1}{\lambda}\right) B_2 a^{-\lambda-2} \right] = a\beta \left[A_2 a^{\lambda-1} + B_2 a^{-\lambda-2} - \frac{k_G}{k_1} \left(A_2 a^{\lambda-1} - \left(1 + \frac{1}{\lambda}\right) B_2 a^{-\lambda-2} \right) \right], \quad (A13)$$

$$A_2 b^{\lambda-1} + B_2 b^{-\lambda-2} = -E, \quad (A14)$$

$$-k_0 E = k_r [A_2 \lambda b^{\lambda-1} - B_2 (\lambda+1) b^{-\lambda-2}]. \quad (A15)$$

These algebraic system can be exactly solved and the constants A_2 and B_2 are derived as

$$A_2^* = \frac{\delta E}{\gamma - \delta}, \quad B_2^* = \frac{-\gamma E}{\gamma - \delta}, \quad (A16)$$

where $A_2^* = A_2 \bullet b^{\lambda-1}$ and $B_2^* = B_2 \bullet b^{-\lambda-2}$. In (A16), the constants δ and γ are defined as

$$\delta = c^{-\frac{\lambda+2}{3}} \bullet \left\{ \frac{a\beta \left[k_1 - \left(1 + \frac{1}{\lambda}\right) k_G \right]}{k_1} + \left(1 + \frac{1}{\lambda}\right) k_G \right\}, \quad (A17)$$

$$\gamma = c^{\frac{\lambda-1}{3}} \bullet \left(k_1 - \lambda k_r + \frac{2\alpha}{a} \right).$$

Now the thermal transparency condition k_0 in (17) can be derived from (A16), (A17) and (A15), and also the temperature field, and the uniform intensity ratio in (22) can be derived from (A16), (A17), (A12) and (26). Lastly, for a composite sphere with HC-type interface, use of (26) in (6) and (A1) will provide 4 conditions

$$A_1 = A_2 a^{\lambda-1} + B_2 a^{-\lambda-2}, \quad (A18)$$

$$k_1 (A_2 a^{\lambda-1} + B_2 a^{-\lambda-2}) - k_G \left[A_2 a^{\lambda-1} - \left(1 + \frac{1}{\lambda}\right) B_2 a^{-\lambda-2} \right] = -\frac{2\alpha}{a} (A_2 a^{\lambda-1} + B_2 a^{-\lambda-2}), \quad (A19)$$

$$A_2 b^{\lambda-1} + B_2 b^{-\lambda-2} = -E, \quad (A20)$$

$$-k_0 E = k_r [A_2 \lambda b^{\lambda-1} - B_2 (\lambda+1) b^{-\lambda-2}], \quad (A21)$$

in which the constants of A_2 and B_2 can be solved as

$$A_2^* = \frac{\delta^* E}{\gamma^* - \delta^*}, \quad B_2^* = \frac{-\gamma^* E}{\gamma^* - \delta^*}, \quad (A22)$$

where A_2^* and B_2^* were defined as in (A16), and now δ^* and γ^* are given by

$$\delta^* = c^{-\frac{\lambda+2}{3}} \bullet \left[k_1 + (\lambda+1) k_r + \frac{2\alpha}{a} \right], \quad \gamma^* = c^{\frac{\lambda-1}{3}} \bullet (k_1 - k_r \lambda + 2\alpha/a). \quad (A23)$$

With the solutions (A22), the thermal transparency condition given in (18) for a composite sphere with HC-type interface can be derived from (A22), (A23) and (A21). While the temperature field, and also the uniform thermal intensity ratio, can be derived from (A22), (A23) and (A18). We mention that the thermal transparency conditions, listed in (8), (9), (17) and (18), can be easily deduced via a series of homogenization procedure as described in Section 3. Detailed graphical illustrations were presented in our earlier works [37,38].

Data availability

Data will be made available on request.

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